

Indrajit Kurmi

Computer Vision Researcher

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Research Interest

My research interests include computational photography, computer vision, and machine learning. My research's primary objective is to discover vision-based solutions for real-world challenges. Over the past few years, I have honed my ability to research, design prototypes, and deploy them quickly. Over the next few years, I hope to become a computer vision scientist who can oversee the development and implementation of comprehensive solutions that can be deployed for real-world scenarios.

Education

JOHANNES KEPLER UNIVERSITY

PHD IN COMPUTER SCIENCE

Thesis submitted Linz, Austria

Specialization:- Computer Vision, Computational Photography

Thesis:- Model-Based Occlusion Removal for Synthetic Aperture Imaging

Advisor:- Prof. Oliver Bimber

I.I.T, KANPUR

M.TECH IN ELECTRICAL ENGINEERING

March 2013 Kanpur, India

Specialization:- Signal processing, Communications and Networks.

Thesis:- Novel Light Field Panorama using Cylindrical Light Field System

Advisor:- Prof. Sumana Gupta and Prof. K. S. Venkatesh

A.D.I.T, GUJARAT

B.TECH IN ELECTRONICS AND COMMUNICATION ENGINEERING

Dec 2010 Anand, India

Skills

PROGRAMMING

Daily Use:

C/C++ Python JavaScript Java Kotlin

Occasional use:

GLSL HTML/CSS XML R

MISCELLANEOUS

Daily Use:

OpenCV OpenGL CUDA LaTeX Git

Occasional use:

Matlab Pytorch OpenCL QT

Experience

GRADUATE RESEARCH ASSISTANT

JOHANNES KEPLER UNIVERSITY

Aug 2017 – Mar 2023

Linz, Austria

- Demonstrated synthetic aperture sensing with drones (airborne optical sectioning – AOS) in applications where occlusion presents a significant hindrance.

C++ Python Cuda OpenGL GLSL SLAM

- Developed an autonomous drone capable of performing search and rescue in forests by deploying a deep learning classification network along with other necessary computations onboard.

Python C++ DNN YOLO Object detection

- Developed an android app for demonstrating AOS capabilities which utilize DJI drones.

Java Kotlin Android DJI-MSDK

- Designed and implemented parallel (camera arrays with drones) and sequential-parallel imaging (drone swarms) systems for detecting moving targets in presence of occlusion such as forests.

Python C++ Anomaly Detection Motion Detection SLAM

RESEARCH ASSISTANT

INDIAN INSTITUTE OF TECHNOLOGY

Mar 2014 – Aug 2017

Kanpur, India

- Pattern Recognition for Anti-counterfeiting system with printed novel tags having 3-D features.

C++ 3D Pattern Recognition

- Implemented artefacts restoration using total variation-based video decomposition.

C++ Optimization Regularization Restoration

- Developed Advanced Flight Data Analysis System for Hindustan Aeronautics Limited.

C++ OpenGL QT Multi Sensor Integration

- Developed Non-Pervasive 3D Wireframe Modeling For Indira Gandhi Center for Atomic Research.

C++ Silhouette segmentation 3D Object Modelling

- Implemented Welding Seam Detection Using SVM For Welspun India.

C++ Matlab ML SVM

SOFTWARE ENGINEER

RFIC FIRMWARE DEVELOPING TEAM, BMOD, ERICSSON GLOBAL INDIA

Sep 2013 – Mar 2014

Bangalore, India

- Involved in Development of the firmware for the RFIC chip.

JOHANNES KEPLER UNIVERSITY

COMPUTER VISION

2019 - 2023 Linz

The course gives insights into image analysis and image understanding for tasks such as object recognition, camera calibration, and structure from motion. In a project, students apply and implement computer vision methods independently.

JOHANNES KEPLER UNIVERSITY

COMPUTER GRAPHICS

2019 - 2023 Linz

Students learn the basic concepts of Computer Graphics such as transformations, the rendering pipeline, scenegraphs, and animations, and practically apply them in an OpenGL/WebGL framework.

I.I.T, KANPUR

IMAGE PROCESSING

Jul 2012 - Dec 2012 Kanpur

Within this course, students learn basic concepts of image processing such as sensing and acquisition, filtering, segmentation, and restoration, and practically apply them in assignments and term projects.

I.I.T, KANPUR

INTRODUCTION TO ELECTRONICS

Jan 2013 - Jun 2013 Kanpur

Students learned a practical introduction to basic circuit design. They demonstrate their learning by designing and building a circuit for their final lab project.

Languages

Novice

Proficient

English

German

Hindi

Gujarati

- **Indrajit Kurmi**, David C. Schedl, and Oliver Bimber. "Combined Person Classification with Airborne Optical Sectioning," *Nature Scientific Reports*, vol. 12, no. 1, pp. 1-11, 2022.
- David C. Schedl, **Indrajit Kurmi**, and Oliver Bimber. "An autonomous drone for search and rescue in forests using airborne optical sectioning," *Science Robotics*, vol. 6, no. 55, pp. eabg1188, 2021.
- David C. Schedl, **Indrajit Kurmi**, and Oliver Bimber, "Search and rescue with airborne optical sectioning," *Nature Machine Intelligence*, vol. 2, no. 12, pp. 783-790, 2020.
- David C. Schedl, **Indrajit Kurmi**, and Oliver Bimber, "Airborne optical sectioning for nesting observation," *Nature Scientific Reports*, vol. 10, no. 1, pp. 1-7, 2020.
- **Indrajit Kurmi**, David C. Schedl, and Oliver Bimber, "Airborne optical sectioning," *Journal of Imaging*, vol. 4, no. 8, pp. 102, 2018.
- Oliver Bimber, **Indrajit Kurmi**, and David C. Schedl, "Synthetic aperture imaging with drones," *IEEE Computer Graphics and Applications*, vol. 39, no. 3, pp. 8-15, 2019.
- **Indrajit Kurmi**, David C. Schedl, and Oliver Bimber, "A statistical view on synthetic aperture imaging for occlusion removal," *IEEE Sensors*, vol. 19, no. 20, pp. 9374-9383, 2019.
- **Indrajit Kurmi**, David C. Schedl, and Oliver Bimber, "Thermal airborne optical sectioning," *Remote Sensing*, vol. 11, no. 14, pp. 1668, 2019.
- **Indrajit Kurmi**, David C. Schedl, and Oliver Bimber, "Fast automatic visibility optimization for thermal synthetic aperture visualization," *IEEE Geoscience and Remote Sensing Letters*, vol. 18, no. 5, pp. 836-840, 2020.
- **Indrajit Kurmi**, David C. Schedl, and Oliver Bimber, "Pose Error Reduction for Focus Enhancement in Thermal Synthetic Aperture Visualization," *IEEE Geoscience and Remote Sensing Letters*, pp. 1-5, doi: 10.1109/LGRS.2021.3051718, 2021.
- Rakesh John Amala Arokia Nathan, **Indrajit Kurmi**, and Oliver Bimber, "Inverse Airborne Optical Sectioning," *Drones*, vol. 6, no. 9, 2022.
- Francis Seits, **Indrajit Kurmi**, and Oliver Bimber, "Evaluation of Color Anomaly Detection in Multispectral Images For Synthetic Aperture Sensing," *Eng*, vol. 3, no. 4, pp. 541-553, 2022.
- Rakesh John Amala Arokia Nathan, **Indrajit Kurmi**, David C. Schedl, and Oliver Bimber, "Through-Foliage Tracking with Airborne Optical Sectioning," *Journal of Remote Sensing*, vol. 2022, no. 9812765, 2022.
- **Indrajit Kurmi**, David C. Schedl, and Oliver Bimber. "Micro-lens aperture array for enhanced thin-film imaging using luminescent concentrators," *Optics Express*, vol. 26, no. 22, pp. 29253-29261, 2018.
- David C. Schedl, **Indrajit Kurmi**, and Oliver Bimber. "Voxelizing Light-Field Recordings," *Eurographics*, posters proceedings, pp. 3-5, 2019.
- Oliver Bimber and **Indrajit Kurmi**, "A 1mm Thin-Film Short-Distance Optical Sensor," in *Imaging and Applied Optics 2019 (COSI, IS, MATH, pcAOP)*, OSA Technical Digest (Optical Society of America, 2019), paper ITh1B.3, 2019.
- Oliver Bimber and **Indrajit Kurmi**, "Towards a one millimeter thin foil camera," *International Journal of Wavelets, Multiresolution and Information Processing*, pp. 2040002, doi:10.1142/S0219691320400020, 2020.
- Oliver Bimber et al., "Autonomous Drone for Search and Rescue in Forests," *Eurographics 2021*, (Projects and Labs), 2021.
- Francis Seits, **Indrajit Kurmi**, Rakesh John Amala Arokia Nathan, Rudolf Ortner, and Oliver Bimber, "On the Role of Field of View for Occlusion Removal with Airborne Optical Sectioning," arxiv :<https://arxiv.org/abs/2204.13371>, 2022.